# Socioeconomic Factors and ACT Performance


## Abstract

The purpose of this project was to look at whether socioeconomic factors can help predict high school performance using average ACT score as the measure. The data used is from EdGap and combined with school information from the National Center for Education Statistics Common Core of Data (CCD). The main socioeconomic variables included are: unemployment rate, percent of adults with a college degree, percent of children in married-couple families, median household income, and percent of students receiving free or reduced-price lunch.

I also examined charter school status as an additional predictor to see whether charter schools had different ACT scores than non-charter schools after socioeconomic factors were taken into account. The regression model explained 60.9% variation in average ACT scores. unemployment, percent college, education and free or reduced lunch were statistically significant predictors. Charter status was not statistically significant after controlling for the socioeconomic variables. The results show that socioeconomic factors are strongly related to differences in ACT performance.


## Introduction

School performance can be affected by more that what occurs in the classroom. Economic and social conditions surrounding the students can also play a role in their academic outcomes. Schools in areas with higher unemployment, lower attainment, or more economic hardships prove to face difficult challenges than schools in economically advantaged areas.

The main goal of the project was to answer the question:

**Are average ACT scores predicted by socioeconomic factors?**

The factors included unemployment rate, percent of adults with a college degree, percent of children living in married-couple families, median household income, and percent of students receiving free or reduced-price lunch.

I also wanted to examine an additional characteristic. My second question was:

**Do charter schools have different average ACT scores than non-chartered schools after the socioeconomic factors were considered?**

I wanted to look at whether charter status added any additional information after taking socioeconomic differences into account.

**Theoretical Background**

Unfortunately, socioeconomic conditions affect students in several ways. Parental income, unemployment, education level, and access to resources can all be related to the opportunities available to students and schools.

For this project, I expected schools in areas with higher levels of adult education tend to generally have higher ACT scores. I also expected higher unemployment and higher percentages of students receiving free or reduced lunch to be associated with lower ACT scores because these variables tend to represent greater economic disadvantage.

Median household income and percentage of children living in married couple families were also included because they potentially can represent additional differences in the economic and family environments surrounding schools.

Charter status was additionally included. Rather than individually comparing the average ACT scores of charter and non-charter schools, I wanted to see whether charter status still matters after the socioeconomic differences between schools were considered.

# Methodology

Data for this analysis came from EdGap and the National Center for Education Statistics Common Core of Data (CCD) for the 2016–2017 school year. The datasets were merged using the NCES school identification number. Average ACT score was used as the outcome variable. The socioeconomic predictors were unemployment rate, percent of adults with a college degree, percent of children living in married-couple families, median household income, and percent of students receiving free or reduced-price lunch. Charter status from the CCD dataset was included as an additional predictor.

Before analysis, the datasets were inspected for incorrect data types, duplicates, missing values, and invalid values. Column names were renamed using lowercase snake_case formatting. Negative values for free or reduced-price lunch and ACT scores below one were treated as missing. Missing values in the socioeconomic predictors were handled using iterative imputation, while missing ACT scores were not imputed. The cleaned dataset was then exported for reproducibility.

Exploratory analysis included descriptive statistics, scatterplots, correlation plots, and residual plots. Ordinary least squares multiple regression was used to predict average ACT score from the five socioeconomic variables. A reduced model containing unemployment, percent college educated, and free or reduced-price lunch was also evaluated.

For the additional analysis, charter status was coded as 0 for non-charter schools and 1 for charter schools. A second regression model included charter status along with all five

socioeconomic predictors to determine whether charter status was related to ACT performance after socioeconomic factors were considered. Statistical significance was evaluated at the 0.05 level.

## Computational Results

The full socioeconomic regression model explained approximately 60.9% of the variation in average ACT scores ($R^2 = 0.609$). Unemployment, percent of adults with a college degree, and free or reduced-price lunch were statistically significant predictors. Unemployment had a negative coefficient of approximately -2.17 ($p < .001$), percent college had a positive coefficient of approximately 1.73 ($p < .001$), and free or reduced-price lunch had a negative coefficient of approximately -7.54 ($p < .001$). Percent married and median household income were not statistically significant. A reduced model using unemployment, percent college, and free or reduced-price lunch produced nearly the same $R^2$, suggesting the other two variables added little predictive value.

For the charter analysis, non-charter schools had an average ACT score of approximately 20.30, compared with 19.06 for charter schools. After controlling for the five socioeconomic predictors, the charter coefficient was approximately -0.17 with a p-value of 0.065. Because this p-value was greater than 0.05, charter status was not statistically significant after socioeconomic factors were taken into account.

## Discussion

The results showed that three of the socioeconomic factors were statistically significant. Percent of adults with a college degree had a positive relationship with average ACT scores, while unemployment and free or reduced-price lunch had negative relationships with ACT scores. Percent married and median household income were not statistically significant once the other variables were included in the model.

For the charter school analysis, charter schools had lower ACT scores on average, but charter status was not statistically significant after controlling for the socioeconomic factors ($p = 0.065$). This suggests that the socioeconomic variables explain more of the differences in ACT scores than charter status by itself.

One limitation of this analysis is that it only shows relationships between the variables and does not prove cause and effect. The data also use school averages instead of individual student-level information.

## Conclusion

The results of this project suggest that socioeconomic factors are important predictors of school ACT performance.

Together, unemployment, adult college education, percent married, median income, and free or reduced-price lunch explained about 60.9% of the variation in average ACT scores.

Unemployment, adult college education, and free or reduced-price lunch were the strongest predictors in the model. Higher college education was associated with higher ACT scores, while unemployment and free or reduced-price lunch were associated with lower ACT scores.

Charter schools had lower ACT scores than non-charter schools when looking at the raw averages. However, after socioeconomic factors were taken into account, charter status was not statistically significant.

Overall, the results suggest that socioeconomic conditions may be more useful for explaining differences in ACT performance than whether a school is charter or non-charter.

## References

EdGap. (n.d.). *EdGap*. https://www.edgap.org/

National Center for Education Statistics. *Common Core of Data (CCD), Public Elementary/Secondary School Universe Survey, 2016–2017*. U.S. Department of Education.